

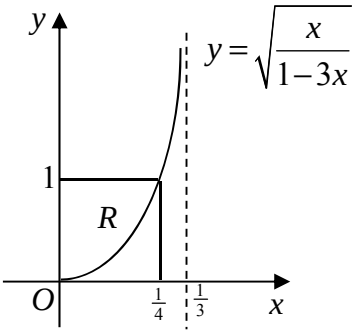


Qn	Solution
1	$f(x) = e^{\sin x}$ $= e^{x - \frac{x^3}{3!} + \dots}$ $= 1 + \left(x - \frac{x^3}{3!}\right) + \frac{\left(x - \frac{x^3}{3!}\right)^2}{2!} + \frac{\left(x - \frac{x^3}{3!}\right)^3}{3!} + \dots$ $= 1 + x - \frac{x^3}{6} + \frac{x^2}{2} + \frac{x^3}{6} + \dots$ $= 1 + x + \frac{x^2}{2} + \dots$ $\therefore a=1, b=1, c=\frac{1}{2} \text{ and } d=0.$
	$\frac{1}{\left(e^{\sin x}\right)^2} \approx \left(1 + x + \frac{x^2}{2}\right)^{-2}$
	$= 1 + (-2)\left(x + \frac{x^2}{2}\right) + \frac{(-2)(-3)}{2!}\left(x + \frac{x^2}{2}\right)^2 + \dots$ $\approx 1 - 2x + 2x^2$
(i)	$4 \frac{dy}{dx} = (y+1)^2$ Differentiating with respect to x, $4 \frac{d^2y}{dx^2} = 2(y+1) \frac{dy}{dx}$ $4 \frac{d^3y}{dx^3} = 2(y+1) \left(\frac{d^2y}{dx^2}\right) + 2\left(\frac{dy}{dx}\right)^2$
	Sub $x=0$, $y=1$, $\frac{dy}{dx}=1$, $\frac{d^2y}{dx^2}=1$, $\frac{d^3y}{dx^3}=\frac{3}{2}$
	Using Maclaurin's formula, $g(x) = 1 + x + \frac{x^2}{2!} + \left(\frac{3}{2}\right) \frac{x^3}{3!} + \dots$
	$g(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{4} + \dots$
	$g(x) - f(x) = \left(1 + x + \frac{x^2}{2} + \frac{x^3}{4} + \dots\right) - \left(1 + x + \frac{1}{2}x^2 + \dots\right)$ $\approx \frac{x^3}{4}$

(ii)	As $x \rightarrow 0$, $g(x) - f(x) \approx \frac{1}{4}x^3 \rightarrow 0$. Therefore, $f(x)$ is a good approximation to $g(x)$ for values of x close to zero.
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2	GP: $a = 10000, r = 1.07$	
	$U_{15} = 10000(1.07)^{14}$	
	$U_{15} = 25785.34 \approx 25785$	
(i)	First three terms of G.P.: $a + 6d, a + 2d, a$ $\Rightarrow \frac{a + 2d}{a + 6d} = \frac{a}{a + 2d}$	
	$a^2 + 4ad + 4d^2 = a^2 + 6ad$ $4d^2 = 2ad$ $d \neq 0 \Rightarrow d = \frac{a}{2}$	
(ii)	Given $a + 6d = 3000$, where $d = \frac{a}{2}$ from (i) $a + 3a = 3000 \Rightarrow a = \frac{3000}{4} = 750$	
	Total calories loss $S_n = \frac{n}{2} [2(750) + (n-1)(375)]$ $= \frac{375n}{2} (3 + n)$	
	G.P. $U_1 = 3000, r = \frac{a}{a + 2\left(\frac{a}{2}\right)} = \frac{1}{2}$ Total calories gain $S_n = \frac{3000 \left(1 - \left(\frac{1}{2}\right)^n\right)}{1 - \frac{1}{2}} = 6000 \left(1 - \left(\frac{1}{2}\right)^n\right)$	
	$\frac{375n}{2} (3 + n) - 6000 \left(1 - \left(\frac{1}{2}\right)^n\right) \geq 200000$ $\frac{3n}{16} (3 + n) - 6 \left(1 - \left(\frac{1}{2}\right)^n\right) \geq 200$	
	From GC, $n \geq 31.68$ (or 32) Least number of weeks = 32.	

3(i)	$\sum_{r=1}^n \frac{1}{r(r+1)} = \sum_{r=1}^n \left[\frac{1}{r} - \frac{1}{(r+1)} \right]$ $= \left\{ \begin{array}{l} \frac{1}{1} - \frac{1}{2} \\ + \frac{1}{2} - \frac{1}{3} \\ + \dots \\ + \frac{1}{n-1} - \frac{1}{n} \\ + \frac{1}{n} - \frac{1}{n+1} \end{array} \right\}$ $= 1 - \frac{1}{n+1}$	
	$\sum_{r=1}^n \left[3^{-r} - \frac{1}{r(r+1)} \right] = \sum_{r=1}^n 3^{-r} - \sum_{r=1}^n \frac{1}{r(r+1)}$ $= \frac{\frac{1}{3} \left(1 - \frac{1}{3^n} \right)}{1 - \frac{1}{3}} - \left(1 - \frac{1}{n+1} \right)$ $= \frac{1}{2} \left(1 - \frac{1}{3^n} \right) - 1 + \frac{1}{n+1}$ $= -\frac{1}{2} \left(\frac{1}{3^n} \right) + \frac{1}{n+1} - \frac{1}{2}$	
(ii)	$\sum_{r=3}^{2N} \left[3^{1-r} - \frac{1}{r(r-1)} \right] = \sum_{r=2}^{2N-1} \left[3^{-r} - \frac{1}{r(r+1)} \right]$ $= \sum_{r=1}^{2N-1} \left[3^{-r} - \frac{1}{r(r+1)} \right] - \left(\frac{1}{3} - \frac{1}{2} \right)$ $= -\frac{1}{2} \left(\frac{1}{3^{2N-1}} \right) + \frac{1}{2N} - \frac{1}{2} + \frac{1}{6}$ $= -\frac{1}{3} + \frac{1}{2} \left(\frac{1}{N} - \frac{1}{3^{2N-1}} \right) \quad [\text{Shown}]$	
	$\sum_{r=3}^{\infty} \left[3^{1-r} - \frac{1}{r(r-1)} \right] = \lim_{N \rightarrow \infty} \left[-\frac{1}{3} + \frac{1}{2} \left(\frac{1}{N} - \frac{1}{3^{2N-1}} \right) \right] = -\frac{1}{3}$	

4(i)	$x = \frac{1}{3} \sin^2 \theta \Rightarrow \frac{dx}{d\theta} = \frac{2}{3} \sin \theta \cos \theta$ When $x = 0$, $\theta = 0$; when $x = \frac{1}{4}$, $\theta = \frac{\pi}{3}$. $\int_0^{\frac{1}{4}} \sqrt{\frac{x}{1-3x}} dx = \int_0^{\frac{\pi}{3}} \sqrt{\frac{\frac{1}{3} \sin^2 \theta}{\cos^2 \theta}} \left(\frac{2}{3} \sin \theta \cos \theta d\theta \right)$ $= \frac{2}{3\sqrt{3}} \int_0^{\frac{\pi}{3}} \sin^2 \theta d\theta$ $= \frac{2}{3\sqrt{3}} \int_0^{\frac{\pi}{3}} \frac{1 - \cos 2\theta}{2} d\theta$ $= \frac{1}{3\sqrt{3}} \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{3}}$ $= \frac{1}{3\sqrt{3}} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$	
(ii)	 Area of $R = \frac{1}{4} - \frac{1}{3\sqrt{3}} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{4} \right]$ $= \frac{1}{3} - \frac{\pi}{9\sqrt{3}}$	
(iii)	$y = \sqrt{\frac{x}{1-3x}} \Rightarrow y^2 = \frac{x}{1-3x}$ $y^2 - 3xy^2 = x$ $x(1+3y^2) = y^2$ $x = \frac{y^2}{1+3y^2}$ Volume = $\pi \int_0^1 \left(\frac{y^2}{1+3y^2} \right)^2 dy$ $= 0.0761 \quad (4 \text{ dp})$	

5(i)	No. of different ways = ${}^{10}C_7 \times {}^{12}C_7$	
	= 95040	
(ii)	No. of teams including Jason and Joyce $= {}^9C_6 \times {}^{11}C_6$	
	Required probability $= \frac{{}^9C_6 \times {}^{11}C_6}{95040}$ $= \frac{38808}{95040}$ $= \frac{49}{120}$ or 0.408	
(iii)	No. of teams including Joel $= {}^9C_6 \times {}^{12}C_7$ $= 66528$	
	No. of teams including Joel and Jason but not Joyce $= {}^8C_5 \times {}^{11}C_7$ $= 18480$ No. of teams including Joel and Joyce but not Jason $= {}^8C_6 \times {}^{11}C_6$ $= 12936$	
	Required probability $= \frac{18480 + 12936}{66528}$	
	$= \frac{17}{36}$ or 0.472	

6(i)	Table of outcomes:																																																							
	<table><tr><td>$X_1 \backslash X_2$</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>1</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>2</td><td>2</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>3</td><td>3</td><td>3</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>4</td><td>4</td><td>4</td><td>4</td><td>4</td><td>5</td><td>6</td></tr><tr><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>5</td><td>6</td></tr><tr><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td><td>6</td></tr></table>	$X_1 \backslash X_2$	1	2	3	4	5	6	1	1	2	3	4	5	6	2	2	2	3	4	5	6	3	3	3	3	4	5	6	4	4	4	4	4	5	6	5	5	5	5	5	5	6	6	6	6	6	6	6	6						
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$P(\text{one number} \leq x) = \frac{x}{6}$																																																								
$P(X \leq x) = P(\text{both numbers} \leq x), \ x = 1, 2, \dots, 6.$ $= P(X_1 = 1, 2, \dots, x \ \& \ X_2 = 1, 2, \dots, x)$ $= \left(\frac{x}{6}\right)\left(\frac{x}{6}\right)$ $= \left(\frac{x}{6}\right)^2$																																																								
(ii)	$P(X = x) = P(X \leq x) - P(X \leq x - 1)$ $= \left(\frac{x}{6}\right)^2 - \left(\frac{x - 1}{6}\right)^2$ $= \frac{2x - 1}{36}$																																																							
(iii)	<table><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$P(X = x)$</td><td>$\frac{1}{36}$</td><td>$\frac{3}{36}$</td><td>$\frac{5}{36}$</td><td>$\frac{7}{36}$</td><td>$\frac{9}{36}$</td><td>$\frac{11}{36}$</td></tr></table>	x	1	2	3	4	5	6	$P(X = x)$	$\frac{1}{36}$	$\frac{3}{36}$	$\frac{5}{36}$	$\frac{7}{36}$	$\frac{9}{36}$	$\frac{11}{36}$																																									
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	$E(X) = \sum_{\text{all } r} xP(X = x)$ $= 1 \times \frac{1}{36} + 2 \times \frac{3}{36} + 3 \times \frac{5}{36} + 4 \times \frac{7}{36} + 5 \times \frac{9}{36} + 6 \times \frac{11}{36} = \frac{161}{36}$ [Shown]																																																							
$E(X^2) = \sum_{\text{all } r} x^2P(X = x)$ $= 1^2 \times \frac{1}{36} + 2^2 \times \frac{3}{36} + 3^2 \times \frac{5}{36} + 4^2 \times \frac{7}{36} + 5^2 \times \frac{9}{36} + 6^2 \times \frac{11}{36}$ $= \frac{791}{36}$																																																								
$\text{Var}(X) = E(X^2) - [E(X)]^2$																																																								

	$= \frac{791}{36} - \left(\frac{161}{36}\right)^2$ $= \frac{2555}{1296} \quad [\text{Shown}]$	
(iv)	Since $n = 40$ is large, by Central Limit Theorem, $\bar{X} \sim N\left(\frac{161}{36}, \frac{2555}{1296 \times 40}\right) \text{ approximately.}$	
	$P(\bar{X} \geq 4.5) = 0.45021 \approx 0.450 \text{ (3 sig figs)}$	

7(i)	<table><caption>Data points from the scatter plot</caption><thead><tr><th>t</th><th>y</th></tr></thead><tbody><tr><td>15</td><td>30</td></tr><tr><td>30</td><td>19</td></tr><tr><td>45</td><td>15</td></tr><tr><td>60</td><td>13</td></tr><tr><td>75</td><td>12</td></tr><tr><td>90</td><td>11</td></tr><tr><td>105</td><td>10</td></tr></tbody></table>	t	y	15	30	30	19	45	15	60	13	75	12	90	11	105	10	
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(ii)	From G.C., (a) <u>$r = -0.8745$</u> (ans) (b) <u>$r = -0.9288$</u> (ans) (c) <u>$r = 0.9993$</u> (ans)																	
	Model (c) is the most appropriate model for the relationship between t and y since its value of $ r $ is closest to 1.																	

(iii)	From G.C., $y = 7.2048 + 344.60 \left(\frac{1}{t} \right)$ $y = 7.20 + 345 \left(\frac{1}{t} \right) \quad (3\text{s.f.})(\text{ans})$	
(iv)	When $t = 120$, $y = 7.2048 + 344.60 \left(\frac{1}{120} \right) = 10.076$ <u>$y ; 10$ (ans)</u> Since $t = 120$ is outside the given range of the values of t, this is an extrapolation and thus the prediction may not be reliable.	
8(i)	It is inappropriate to model the number of blue marbles by a binomial distribution because the marbles are drawn without replacement, the colour of the marbles depends on that of the previous draw.	
(ii)	Required probability $= \frac{10}{18} \times \frac{9}{17} \times \frac{8}{16} \times \frac{8}{15} \times \frac{7}{14} \times \frac{5!}{3! \times 2!} \text{ or } \frac{{}^{10}C_3 \times {}^8C_2}{{}^{18}C_5}$ $= \frac{20}{51}$ or 0.392	
(iii)	$R \sim B\left(n, \frac{12}{32}\right) = B\left(n, \frac{3}{8}\right)$ $E(R) = 4.5$ $\Rightarrow \frac{3}{8}n = 4.5$ $n = 12$ $P(R > 4) = 1 - P(R \leq 4)$ $= 0.48972 \approx 0.490$ (3 sig figs)	
(iv)	$R \sim B\left(n, \frac{3}{8}\right)$ $P(R = 0 \text{ or } 1) < 0.01$ $\Rightarrow P(R = 0) + P(R = 1) < 0.01$ $\Rightarrow \left(\frac{5}{8}\right)^n + n\left(\frac{3}{8}\right)\left(\frac{5}{8}\right)^{n-1} < 0.01$	
	From GC, least $n = 15$	

9(i)	Let D = Mass of a durian and R = Mass of a melon. Then $D \sim N(2.1, 0.25^2)$ and $R \sim N(0.6, 0.16^2)$. $D - 4R \sim N(2.1 - 4(0.6), 0.25^2 + 4^2 \times 0.16^2)$	
	$D - 4R \sim N(-0.3, 0.4721)$	
	$P(D - 4R < 0) \approx 0.66881 = 0.669$ (to 3 sig figs)	
(ii)	Let $M = \frac{D_1 + D_2 + R_1 + R_2 + L + R_8}{10}$	
	$E(M) = \frac{1}{10} [2(2.1) + 8(0.6)] = 0.9$	
	$\text{Var}(M) = \frac{1}{10^2} [2(0.25)^2 + 8(0.16)^2] = 0.003298$	
	$M \sim N(0.9, 0.003298)$	
	$P(M > 1) = 0.040815 = 0.0408$ (to 3 sig figs)	
(iii)	Let $S = 15D + 6R$	
	$15D \sim N(15(2.1), 15^2 \times 0.25^2)$ and $6R \sim N(6(0.6), 6^2 \times 0.16^2)$	
	$S \sim N(31.5 + 3.6, 14.0625 + 0.9216) = N(35.1, 14.9841)$	
	$P(S < 40) = 0.89722 = 0.897$	
(iv)	Event in part (iii) includes the event in part (iv) plus some other cases. [For example, the case where $15D < 33$ and $6R < 7$ is included in (iii) but not in (iv) .]	

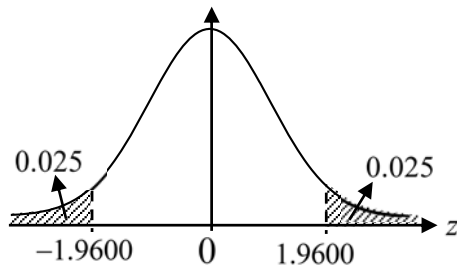
10(i)	A 1-tail test should be used because he is investigating for an average time longer than 60 minutes	
(ii)	Since the sample size is large, the public relation officer can apply the Central Limit Theorem to approximate the distribution of the sample mean ($\bar{X}$) by a normal distribution to conduct a hypothesis test.	
(iii)	unbiased estimate of population mean $= \bar{x}$ $= 50 + \frac{1071}{70}$ $= \frac{653}{10}$	
	Unbiased estimate of population variance $= s^2$ $= \frac{1}{69} \left(73158 - \frac{1071^2}{70} \right)$ $= \frac{189239}{230}$	
(iv)	Let μ be the population mean of X . $H_0 : \mu = 60$ $H_1 : \mu > 60$	
	Under H_0 , since $n = 70$ is large, by the Central Limit Theorem, $\bar{X} : N \left(60, \frac{189239}{230(70)} \right)$ approximately.	
	Test Statistic : $Z = \frac{\bar{X} - 60}{\sqrt{\frac{189239}{230(70)}}} : N(0,1)$ approximately	
	Level of significance: 5% By using G.C., $p - \text{value} = 0.0611$ (3 s.f)	
	Since $p\text{-value} = 0.0611 > 0.05$, we do not reject H_0 at the 5% level of significance and conclude that there is insufficient evidence that the population mean waiting time is longer than 60 minutes. i.e. The complaint is not valid.	

(v)

$$H_0 : \mu = k$$

$$H_1 : \mu \neq k$$

Level of significance: 5%



For H_0 to be rejected,

$$z \leq -1.9600 \text{ or } z \geq 1.9600$$

$$\begin{aligned} \frac{\bar{x} - k}{s/\sqrt{70}} &\leq -1.9600 \text{ or } \frac{\bar{x} - k}{s/\sqrt{70}} \geq 1.9600 \\ \frac{65.3 - k}{\sqrt{822.778/70}} &\leq -1.9600 \text{ or } \frac{65.3 - k}{\sqrt{822.778/70}} \geq 1.9600 \\ 65.3 - k &\leq -6.7197 \text{ or } 65.3 - k \geq 6.7197 \end{aligned}$$

$$\Rightarrow \underline{\underline{k \leq 58.6 \text{ or } k \geq 72.0 \text{ (ans)}}}$$